LLM reads, code verifies

Can you trust what an LLM reads? When should it call the cloud? What if the data can't leave?

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Portfolio & resume

How it works & code

From production

- **90%+** less reconciliation time · sole engineer · 729 PRs
- **7 LLMs** benchmarked on a hand-built 90-receipt, 8-currency set
- **87 s → 10 s** worst-case latency with Gemini → Claude failover
- **Rolled back** confidence-based auto-approval: why this harness exists

1 Can you trust what an LLM reads?

Not its confidence. Check every answer with code.

Four checks on every model answer (no answer key, no model confidence)

Sanitizer: injection patterns in EN/KO/JA/ZH, field whitelist

Arithmetic: subtotal + tax + tip + discount = total

Date plausibility: a real date in the receipts' era

Two-reader: the total and date must appear in the OCR text

Zero flags → **auto-accept**. Flagged → escalate one rung → still flagged → **human review**.

With no answer key, the checks caught

38 of 51
wrong answers from the local model

5 of 8
wrong answers from the cloud model

Adversarial set (cloud, cheap)	n	model fooled	gate fired	leaked
Date traps	10	0	10	0
Arithmetic traps	11	1	11	0
Injection, blatant	6	0	6	0
Injection, disguised	10	0	10	0
Controls	6	0	6	0

2 When should it call the cloud?

Local first, cloud when the checks fail, people last.

On this machine | Cloud

T1 · OCR + rules	T2 · local VLM	T3 · cloud, cheap	T4 · cloud, strong
RapidOCR + Tesseract · CPU · $0	Qwen2.5-VL 3B · 4 GB GPU · $0	Gemini 3.1 Flash-Lite · $0.54 per 1k	Gemini 3.6 Flash · $1.42 per 1k · Claude failover

Each reader alone	scans: date + total	phone photos	median
Local OCR + rules	72.7%	55.0%	4.7 s
Local VLM 3B	83.0%	81.0%	28.3 s
Cloud, cheap	97.3%	86.5%	1.8 s
Cloud, strong	98.0%	86.5%	1.9 s

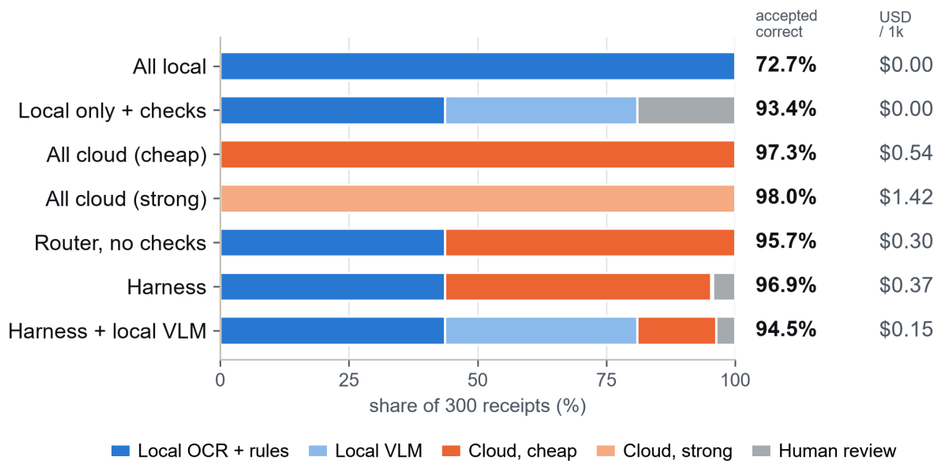


Fig A. Where 300 scans end up under each policy, accepted-answer accuracy and USD per 1,000.

The harness: **96.9% correct** with 56% of receipts sent to the cloud and 4.3% to people, $0.37 per 1,000 (all cloud: 97.3%, $0.54). The strong model adds 0.7 points for 2.7× the price: **escalate to people, not to a bigger model.**

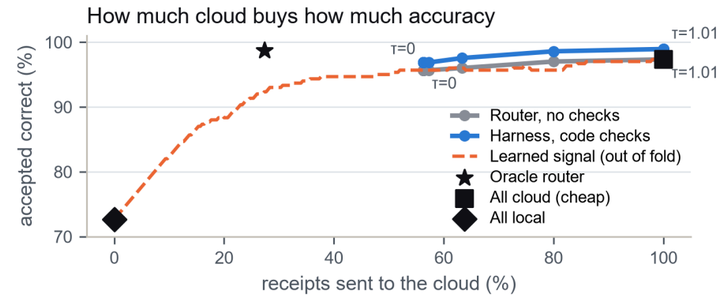


Fig B. Share of receipts sent to the cloud vs accuracy as the hand-off threshold moves; ★ = a router that knew the answers.

A router that knew the answers would send only 27% to the cloud for 98.7%; the simple rules send 56% for 95.7%. **The hard part is knowing when to call the cloud, not the model.** A learned "call the cloud?" score (14 features, out-of-fold AUC 0.89) did no better, so the two simple rules stayed.

3 What if the data can't leave?

Run local-only: same checks, more human review.

	Cloud harness	Local only
Scans: correct	**96.9%**	**93.4%**
Scans: to people	4.3%	19.0%
Scans: USD per 1k	$0.37	$0
Photos: correct	**88.5%**	**89.1%**
Photos: to people	21.5%	31.5%

Photos: same accuracy, more review. Scans: local only gives up 3.4 points. **Keeping data in costs mostly people's time.** In cloud mode the image leaves the machine at one step only; every check and routing decision runs locally.

Check rules are domain assets: a rule built from scan errors caught 10 of 16 there with no false alarms, but 1 receipt on held-out photos, with 1 false alarm.

Next: 10k receipts; invoices and statements; layout features for the hand-off decision; a weekly drift metric.

300 scanned receipts (ICDAR 2019 SROIE) · 200 phone photos (WildReceipt) · 43 synthetic adversarial receipts. Date and total must match exactly; scoring is code, never a model.

Run of 2026-10-31 · code and every result in the repository

Sources. ICDAR 2019 SROIE (Huang et al., arXiv:2103.10213), CC-BY-4.0 · WildReceipt (Sun et al., arXiv:2103.14470) · Tesseract 5.4, RapidOCR / PP-OCR, Qwen2.5-VL (Apache 2.0) via Ollama · Google Gemini API pricing page, 2026-10-31 · Fallback chain and sanitizer adapted from the author's open-source boda (MIT). Production figures are from the author's 2026 engagement; the platform is private.